Physical chemistry cume 12-02-2016

The standard reaction Gibbs energy of the reversible ammonia synthesis reaction

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

is ΔG_r = -32.90 kJ/mol at 298 K. The reaction enthalpy change is ΔH_r = -92.22 kJ/mol.

(a) Calculate the partial pressures of N_2 and H_2 in bar units at equilibrium when pure ammonia gas at 10 bar pressure dissociates slightly at 298 K temperature. (30 p)

(b) Using the Gibbs-Helmholtz equation

$$\frac{\partial}{\partial T}\left(\frac{G}{T}\right) = -\frac{H}{T^2}$$

calculate the standard reaction Gibbs energy at 700 K (assume that ΔH_r is essentially constant between 298 and 700 K). (40 p)

(c) From this, calculate the equilibrium partial pressure of ammonia in a 1:3 mixture of N_2 and H_2 at 700 K and 1 bar total pressure. (30 p)

Constants and notes:

1. R=8.3145 J/(mol K)

2. In (a), the equilibrium is shifted toward ammonia, and therefore expect low partial pressures for N_2 and H_2. The partial pressure of NH_3 can be taken as the initial pressure. Note that the standard free energies for gases correspond to the standard pressure $p_\emptyset$ (1 bar). Assume ideal gas behavior.

3. In (b), integrating the Gibbs-Helmholtz equation from T_1 (298 K) to T_2 (700 K) gives the change of (G/T) from T_1 to T_2. The Gibbs-Helmholtz equation is also valid for free energy *differences*, just use $(\Delta G_r/T)$ and ΔH_r instead of (G/T) and H. Integrating the right-hand side is simple, the indefinite integral is $\int T^{-2}\, dT = -T^{-1}$ + Constant.

4. In part (c), the equilibrium at high temperature is shifted strongly toward N_2 and H_2, and only little N_2 and H_2 is converted to ammonia. Therefore, the calculations can be simplified by assuming that the partial pressures of N_2 and H_2 are 0.25 and 0.75 bar, respectively.

5. **Carry the units on all intermediate quantities. Be careful to distinguish kJ and J.**

6. 1 bar = 10^5 Pa (close to 1 atm) is the convention for the standard pressure $p_\emptyset$.